

Drake Rundell

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EXPERIENCE

General Motors | Full-Stack Software/System Engineer - SDV Embedded Platforms | *WFH* Feb. 2023 – Present

- Built a React/TypeScript + Flask (Python) web application from the ground up — now used by ~200 engineers daily to visualize and navigate SDV software and system architecture; own the full delivery cycle (requirements, design, implementation, deployment, ops) on a 3-person team with no dedicated QA or ops support.
- Led technical evaluation of the graph visualization stack under ambiguous requirements — assessed open-source and commercial options across performance, API ergonomics, and integration complexity; drove the adoption decision and shipped to production.
- Designed multi-source Python ingestion pipeline pulling from planning systems, test result databases, and systems engineering model sources; normalized heterogeneous schemas and unified into PostgreSQL and Neo4j via Databricks on Azure.
- Own automated nightly data pipelines and cloud infrastructure deployment on Azure; serve as internal graph database architect and primary technical point of contact for teams integrating with the platform.

General Motors | Design Release Engineer - Air Delivery | *Warren, MI* July 2021 – Feb. 2023

- First-named author on two granted patents (*US Patent No. 11,933,256* & *11,852,109*) concerning novel approaches to vehicle gas distribution in intake manifold runners — filed independently of any expectation of the role.
- Owned design release for intake manifolds and engine covers across production and development programs; managed 3 suppliers and drove cross-functional reviews spanning design, manufacturing, and validation.

Williams International | Advanced Manufacturing Engineer | *Pontiac, MI* June 2020 – July 2021

- Developed tooling, fixtures, and additive manufacturing solutions for precision electromechanical assemblies; led development of a plastic additive area for rapid, low-cost tooling production.

PROJECTS

Goal Siren — IoT NHL Goal Alert System | *Python, Flask, React, TypeScript, C++, ESP32, Docker, MQTT*

- Designed and built an end-to-end IoT system solo — custom KiCAD PCB, ESP32 C++ firmware with interrupt-driven MQTT, Flask REST API (40+ endpoints), React/TypeScript frontend with MUI, and Docker-containerized backend on a Raspberry Pi Zero; most of these technologies were new when the project started.
- Real-time NHL API integration with <3-second end-to-end latency from API event to physical hardware response via Mosquitto MQTT broker; dynamic PNG card generation via Pillow; SQLite persistence for full CRUD.

RPI Matrix — NHL LED Scoreboard | *TypeScript, React, Node.js, C++, MicroPython, SQLite, Vite, MQTT*

- Full-stack TypeScript system built solo: Node.js backend, React 19 + Vite frontend, MicroPython firmware on ESP32 over MQTT, and native C++ canvas bindings (@napi-rs/canvas) for sub-100ms RGB frame generation streamed to dual HUB75 LED panels; 3,500+ lines TypeScript end to end.
- Server-Sent Events for real-time browser preview without polling; async display state machine with AbortController; drag-and-drop ordering with @dnd-kit; data-heavy dashboard across SQLite with 11 tables.

PlantNode — Environmental Monitor | *C++, Python, ESP32-C3, KiCAD, LVGL, I2C, UART*

- Designed custom 2-layer KiCAD PCB and Fusion 360 enclosure for an ESP32-C3 sensor node; multi-sensor array (ADC, time-of-flight, lux, soil moisture) over I2C with 8-byte UART telemetry protocol and XOR checksum.
- Parametric C++ rendering engine (LVGL v9, <16ms frame budget, rect primitives with per-leaf sway animation); Python tooling mirrors firmware geometry for desktop preview and automated C/Python stage-table sync.

EDUCATION

Georgia Institute of Technology - OMSCS August 2022 – Present

MSE Computer Science - Computing Systems

- **Relevant coursework:** *Graduate Operating Systems, Information Security, Computer Networks, Machine Learning for Trading*

University of Michigan - Ann Arbor Grad. May 2020

BSE Aerospace Engineering